project_part_2_Team_#

April 25, 2026

1 Project Part 2: Data Understanding and Preparation

- **Team number:** `#`
- **Team members:** `Cole Beach`, `Eric Lindquist`, `Tuan Lo`
- **Dataset name:** `Music Genre Classification Dataset (train.csv / test.csv)`
- **Target variable:** `Class`

1.1 1. Problem and Data Snapshot

- **Dataset:** We use the music classification dataset from `data/raw/train.csv` for training and `data/raw/test.csv` for prediction.
- **Machine learning problem:** This is a supervised multiclass classification problem because each song has a labeled category.
- **Target variable:** The output to predict is `Class`.
- **Main input features expected to be used:**
  - Metadata features: `Artist Name`, `Track Name`, `Popularity`
  - Audio features: `danceability`, `energy`, `key`, `loudness`, `mode`, `speechiness`, `acousticness`, `instrumentalness`, `liveness`, `valence`, `tempo`, `duration_in min/ms`, `time_signature`

This project aims to learn patterns from track metadata and acoustic descriptors that are useful for predicting the correct music class.

1.2 2. Exploratory Data Analysis

This EDA uses a selective set of variables chosen for modeling relevance, distribution behavior, missingness, and relationship strength.

Dataset snapshot first: we verify overall size (`df.shape`) and schema (`df.info()` style view) so preprocessing decisions are grounded in sample size, numeric coverage, and missingness concentration. The data has enough observations for multiclass modeling, while missingness is concentrated in only a few features rather than spread uniformly.

Focused variables: `Popularity`, `energy`, `loudness`, `acousticness`, `instrumentalness`, `duration_in min/ms`, `speechiness`, `tempo` (target: `Class`).

1.2.1 2.1. EDA Context Snapshot

This subsection standardizes reusable constants, plotting style, and a compact data snapshot.

The snapshot outputs are used to anchor later choices: - Dataset size supports stratified validation without collapsing minority classes. - Missingness concentration (instead of broad random gaps) supports targeted imputation over row dropping. - The descriptive statistics and skew metrics identify where robust scaling and transform sensitivity checks are most important (`speechiness`, `acousticness`, `duration_in min/ms`, `instrumentalness`).

[38]:
```
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns

# ========================
# Configuration / Constants
# ========================
RANDOM_STATE = 42

DATA_DIR = Path("../data/raw")
TRAIN_PATH = DATA_DIR / "train.csv"
TEST_PATH = DATA_DIR / "test.csv"

TARGET_COL = "Class"

FOCUSED_FEATURES = [
    "Popularity",
    "energy",
    "loudness",
    "acousticness",
    "instrumentalness",
    "duration_in min/ms",
    "speechiness",
    "tempo",
]

CATEGORICAL_SNAPSHOT_COLS = ["mode", "time_signature"]
BOUNDED_FEATURES_01 = [
    "danceability",
    "energy",
    "speechiness",
    "acousticness",
    "instrumentalness",
    "liveness",
    "valence",
]

PALETTE = "Set2"
```

```python
CLASS_PALETTE = "tab20"
FIGSIZE_WIDE = (20, 9)
FIGSIZE_MED = (12, 6)
FIGSIZE_HEATMAP = (10, 7)
FIGSIZE_THREE_PANEL = (19, 5)
DPI = 110

sns.set_theme(style="whitegrid", context="notebook", palette=PALETTE)
plt.rcParams["figure.dpi"] = DPI

# Data loading and reusable views.
df = pd.read_csv(TRAIN_PATH)
eda_df = df[FOCUSED_FEATURES + [TARGET_COL]].copy()

# Stable ordering and color mapping for class-aware plots.
class_order = sorted(df[TARGET_COL].dropna().unique().tolist())
class_palette_map = dict(zip(class_order, sns.color_palette(CLASS_PALETTE,␣
 ↪n_colors=len(class_order))))

# Compact snapshot before focused diagnostics.
print("Dataset shape (rows, cols):", df.shape)
print("Target column:", TARGET_COL)

schema_snapshot = pd.DataFrame({
    "dtype": df.dtypes.astype(str),
    "missing_pct": (df.isna().mean() * 100).round(3),
})

print("\nTop columns by missingness:")

audit = pd.DataFrame({
    "dtype": eda_df[FOCUSED_FEATURES].dtypes.astype(str),
    "missing_pct": (eda_df[FOCUSED_FEATURES].isna().mean() * 100).round(3),
    "n_unique": eda_df[FOCUSED_FEATURES].nunique(dropna=True),
}).sort_values("missing_pct", ascending=False)

audit
```

```
Dataset shape (rows, cols): (17996, 17)
Target column: Class

Top columns by missingness:
```

[38]:

	dtype	missing_pct	n_unique
instrumentalness	float64	24.322	4253
Popularity	float64	2.378	100
loudness	float64	0.000	9002

```
energy               float64        0.000      1208
acousticness         float64        0.000      3930
duration_in min/ms   float64        0.000     14121
speechiness          float64        0.000      1194
tempo                float64        0.000     13562
```

[39]:
```python
summary = eda_df[FOCUSED_FEATURES].describe(percentiles=[0.25, 0.5, 0.75]).T
summary = summary.rename(columns={"50%": "median", "25%": "q1", "75%": "q3"})
summary["iqr"] = summary["q3"] - summary["q1"]
summary["missing_pct"] = eda_df[FOCUSED_FEATURES].isna().mean() * 100
summary["skew"] = eda_df[FOCUSED_FEATURES].skew(numeric_only=True)

summary_table = summary[[
    "count",
    "missing_pct",
    "mean",
    "std",
    "min",
    "median",
    "max",
    "iqr",
    "skew",
]].round(3)

summary_table
```

[39]:
```
                      count  missing_pct        mean         std     min  \
Popularity          17568.0        2.378      44.512      17.427   1.000
energy              17996.0        0.000       0.663       0.235   0.000
loudness            17996.0        0.000      -7.911       4.049 -39.952
acousticness        17996.0        0.000       0.247       0.311   0.000
instrumentalness    13619.0       24.322       0.178       0.304   0.000
duration_in min/ms  17996.0        0.000  200744.459  111989.127   0.502
speechiness         17996.0        0.000       0.080       0.084   0.022
tempo               17996.0        0.000     122.623      29.572  30.557

                        median          max        iqr   skew
Popularity              44.000      100.000     23.000  0.076
energy                   0.700        1.000      0.351 -0.661
loudness                -7.016        1.355      4.349 -1.761
acousticness             0.081        0.996      0.430  1.105
instrumentalness         0.004        0.996      0.200  1.531
duration_in min/ms  209160.000  1477187.000  86153.000  0.847
speechiness              0.047        0.955      0.048  3.088
tempo                  120.066      217.416     42.348  0.380
```

- `instrumentalness` has substantial missingness relative to other focused features, which justifies explicit imputation policy testing.

- Multiple features show non-trivial skew (`speechiness`, `acousticness`, `loudness`, `duration_in min/ms`), which argues for robust scaling and possible transform ablations.
- Very different feature scales reinforce using a pipeline to prevent scale-dominated behavior in linear or distance-based models.

1.2.2 2.2. Label and Categorical Sanity Snapshot

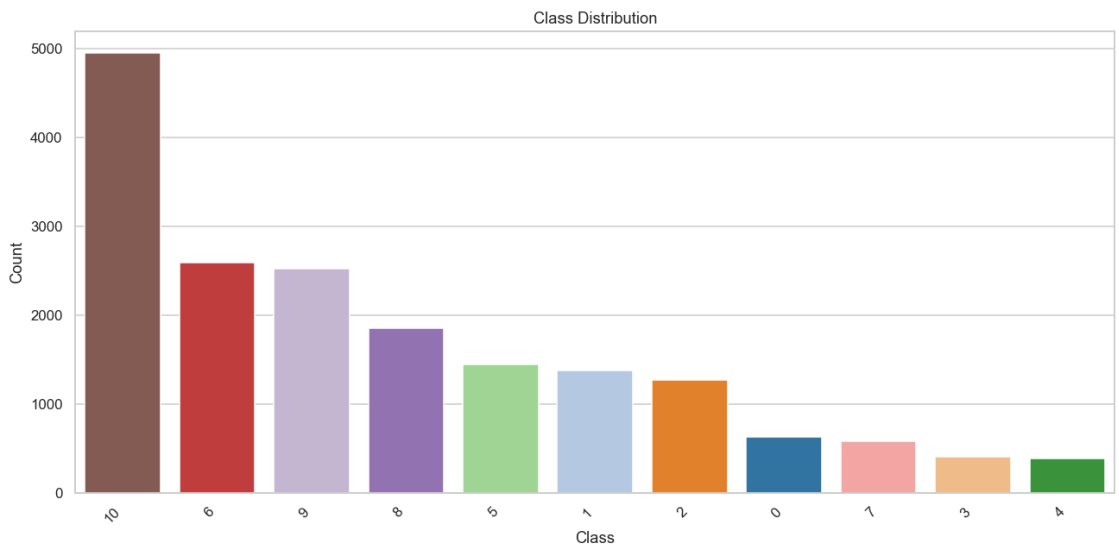
This compact snapshot carries forward the strongest context before focused feature diagnostics.

Interpretation for modeling: - Class counts and percentages confirm moderate imbalance, so later evaluation should emphasize macro-F1 / balanced accuracy rather than only overall accuracy. - Minority classes should be preserved in every split, so stratified train/validation folds are required. - Class-frequency imbalance also motivates testing class weights before more aggressive rebalancing methods.

[40]:
```
# Class distribution (counts + percentages) to support imbalance-aware modeling␣
↪choices.
class_counts = df[TARGET_COL].value_counts().sort_values(ascending=False)
class_pct = (class_counts / class_counts.sum() * 100).round(2)

class_distribution = pd.DataFrame({
    "count": class_counts,
    "pct": class_pct,
})

display(class_distribution)

class_distribution_plot = class_distribution.reset_index().
↪rename(columns={"index": TARGET_COL})

plt.figure(figsize=FIGSIZE_MED)
sns.barplot(
    data=class_distribution_plot,
    x=TARGET_COL,
    y="count",
    order=class_distribution.index.tolist(),
    hue=TARGET_COL,
    palette=class_palette_map,
    legend=False,
)
plt.title("Class Distribution")
plt.xlabel(TARGET_COL)
plt.ylabel("Count")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```

```
       count    pct
Class
```

```
10       4949  27.50
6        2587  14.38
9        2524  14.03
8        1854  10.30
5        1447   8.04
1        1373   7.63
2        1272   7.07
0         625   3.47
7         576   3.20
3         402   2.23
4         387   2.15
```



Why these class-distribution outputs matter: The table and bar chart show that several classes are much smaller than the dominant class, so we should avoid random non-stratified splitting, track per-class errors, and prioritize macro-averaged metrics. This also supports trying class-weighted baselines before synthetic oversampling to reduce overfitting risk on minority classes.

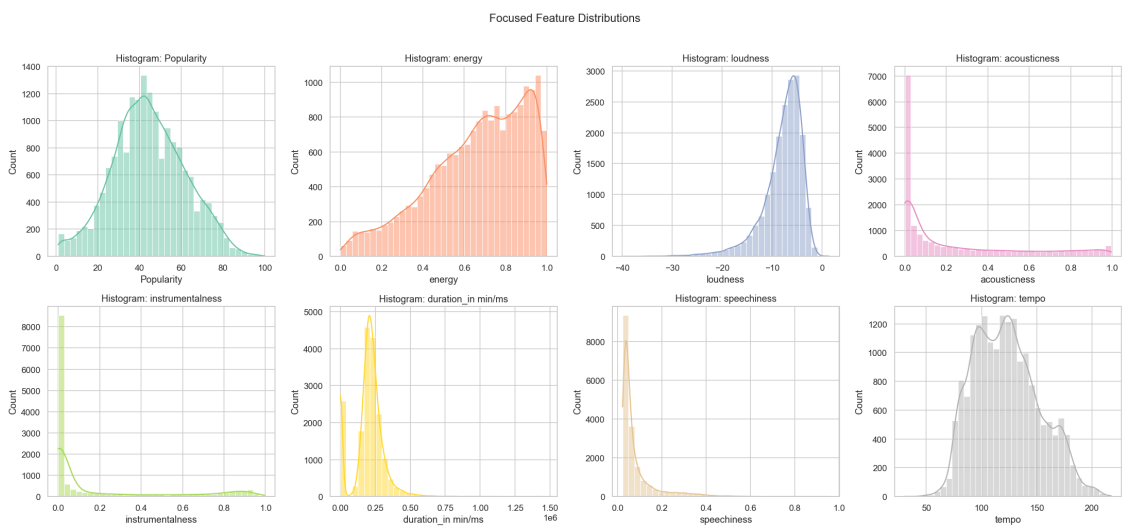
[57]:
```
# Quick categorical sanity checks.
for col in CATEGORICAL_SNAPSHOT_COLS:
    if col in df.columns:
        print(f"\nValue counts for {col}:")
        display(df[col].value_counts(dropna=False).to_frame("count"))
```

```
Value counts for mode:

      count
mode
1     11459
0      6537
```

```
Value counts for time_signature:

                 count
time_signature
4                16451
3                 1228
5                  203
1                  114
```

1.2.3 2.3. Focused Feature Distributions (Histogram)

[42]:
```
# Question: What are the distribution shapes and skew patterns of focused␣
↪features?
plot_df = eda_df[FOCUSED_FEATURES].copy()
feature_palette = sns.color_palette(PALETTE, n_colors=len(FOCUSED_FEATURES))

fig, axes = plt.subplots(2, 4, figsize=FIGSIZE_WIDE)
for i, col in enumerate(FOCUSED_FEATURES):
    ax = axes[i // 4, i % 4]
    sns.histplot(plot_df[col], bins=35, kde=True, ax=ax,␣
↪color=feature_palette[i])
    ax.set_title(f"Histogram: {col}")
    ax.set_xlabel(col)

plt.suptitle("Focused Feature Distributions", y=1.02)
plt.tight_layout()
plt.show()
```



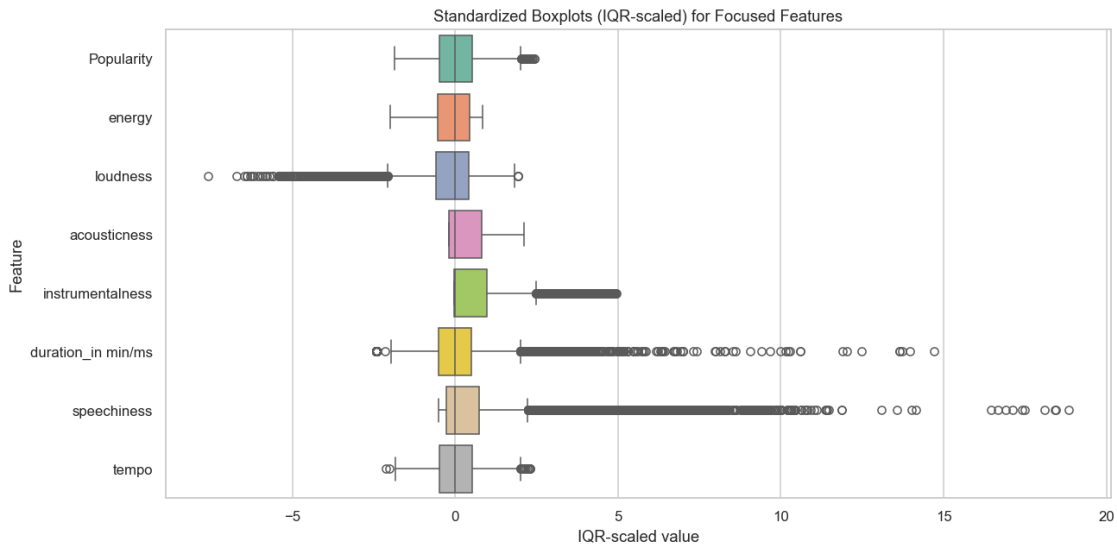
- Histogram shape confirms that `speechiness`, `acousticness`, and `duration_in min/ms` are right-skewed (positive skew means a long right tail: most tracks sit at lower values with a

smaller set of extreme high values).

- This skewness matters because tails can pull linear decision boundaries and distance metrics; robust scaling is preferred over standard scaling for these features, and `log1p` is a reasonable sensitivity check for heavy-tail variables.
- `instrumentalness` has both high missingness and a mass near low values, suggesting many vocal-dominant tracks with a minority of highly instrumental songs. Keep it, impute it, and add a missing-indicator so models can capture both value signal and missingness pattern.
- `duration_in min/ms` has wide spread and clear outliers; tree ensembles usually tolerate this directly, while linear or margin-based models should use clipping/transforms to reduce instability.

1.2.4 2.4. Outlier and Tail Diagnostics (Boxplots)

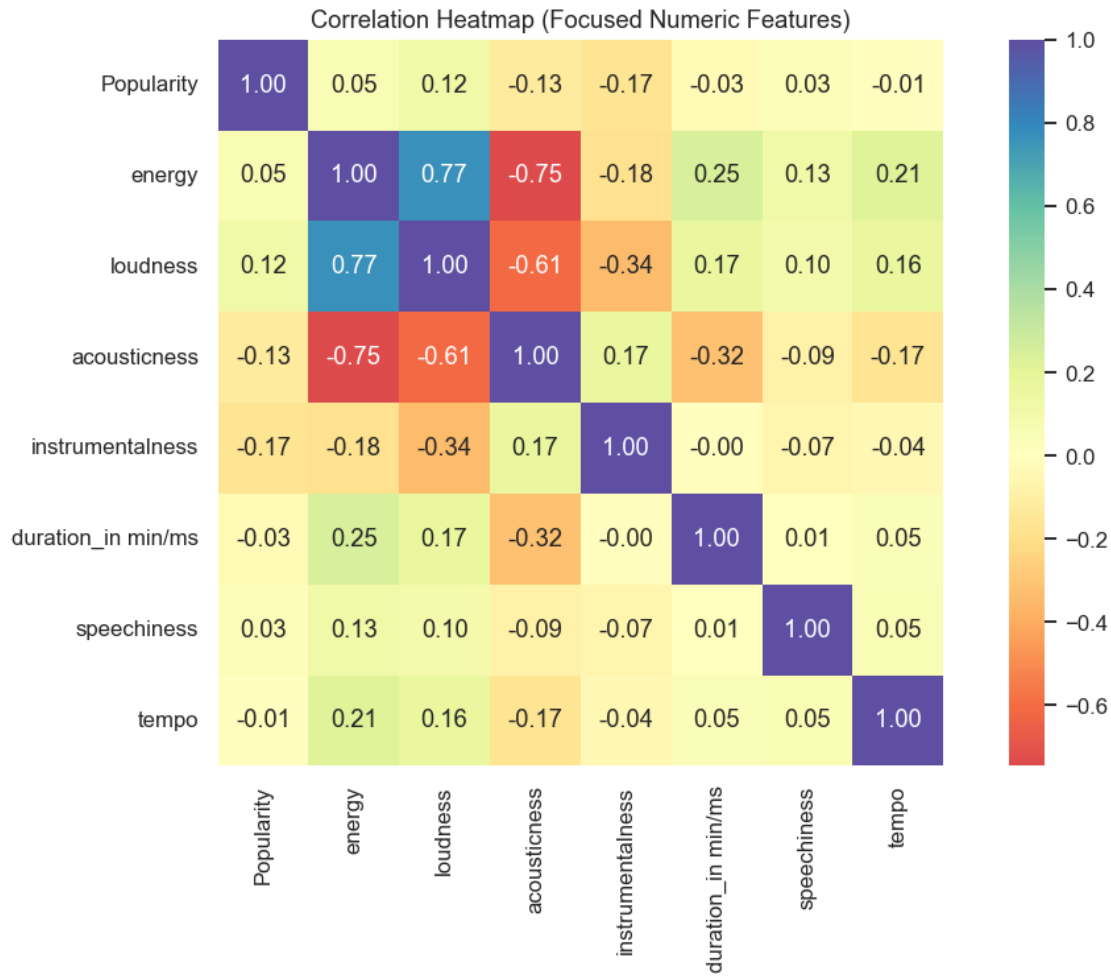
[43]:
```
# Question: Which focused features have heavy tails or potential outliers?
long_df = eda_df[FOCUSED_FEATURES].melt(var_name="feature", value_name="value")
feature_palette = sns.color_palette(PALETTE, n_colors=len(FOCUSED_FEATURES))
feature_palette_map = dict(zip(FOCUSED_FEATURES, feature_palette))

# Standardize within feature so spread is comparable in one compact chart.
long_df["z_value"] = long_df.groupby("feature")["value"].transform(
    lambda x: (x - x.median()) / (x.quantile(0.75) - x.quantile(0.25) + 1e-9)
)

plt.figure(figsize=FIGSIZE_MED)
sns.boxplot(
    data=long_df,
    x="z_value",
    y="feature",
    orient="h",
    whis=1.5,
    hue="feature",
    palette=feature_palette_map,
    legend=False,
)
plt.title("Standardized Boxplots (IQR-scaled) for Focused Features")
plt.xlabel("IQR-scaled value")
plt.ylabel("Feature")
plt.tight_layout()
plt.show()
```



- The standardized boxplots reinforce a heavy-tail profile in several focused variables: `duration_in min/ms`, `speechiness`, and `acousticness` show broader IQR-scaled spread with more extreme points than relatively tighter features.
- `duration_in min/ms` stands out as the most outlier-prone feature, which supports using robust preprocessing (robust scaling and optional clipping/transform sensitivity checks) before linear or distance-based models.
- Outlier density differs by feature rather than appearing uniformly, so a feature-specific treatment strategy is preferable to global row-level outlier removal.
- Because many extreme values in music features can be genuine genre/style signal (not pure noise), keep these observations in baseline models and validate any clipping/removal via cross-validated ablations.

1.2.5 2.5. Correlation Structure of Focused Features

[44]:
```
# Question: Which focused features move together strongly?
corr = eda_df[FOCUSED_FEATURES].corr()

plt.figure(figsize=FIGSIZE_HEATMAP)
sns.heatmap(corr, annot=True, fmt=".2f", cmap="Spectral", center=0, square=True)
plt.title("Correlation Heatmap (Focused Numeric Features)")
plt.tight_layout()
plt.show()
```

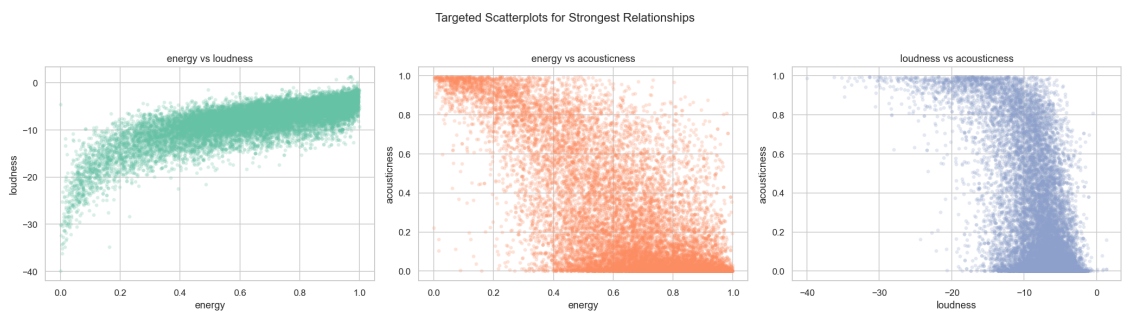


- The heatmap indicates a clear correlation cluster among `energy`, `loudness`, and `acousticness`, consistent with overlapping musical intensity/timbre structure in the feature space.
- This level of collinearity can destabilize coefficient estimates in unregularized linear models, so regularization (or feature-drop ablations) is important for interpretability and generalization.
- Correlation here should be treated as a redundancy signal, not automatic deletion logic: keep the cluster in baseline runs and verify value-add by model family (linear vs tree-based).
- Because `Class` is categorical, this matrix is useful for feature-feature dependency diagnostics only; class relevance should continue to be judged with class-aware methods (ANOVA/MI and class-conditional plots).

1.2.6 2.6. Strongest Pairwise Relationships (Scatterplots)

[45]:
```
# Question: How do the strongest correlated feature pairs behave?
scatter_pairs = [
    ("energy", "loudness"),
    ("energy", "acousticness"),
```

```
    ("loudness", "acousticness"),
]
scatter_palette = sns.color_palette(PALETTE, n_colors=len(scatter_pairs))

fig, axes = plt.subplots(1, 3, figsize=FIGSIZE_THREE_PANEL)
for ax, (x_col, y_col), pair_color in zip(axes, scatter_pairs, scatter_palette):
    sns.scatterplot(
        data=eda_df,
        x=x_col,
        y=y_col,
        alpha=0.25,
        s=18,
        linewidth=0,
        ax=ax,
        color=pair_color,
    )
    ax.set_title(f"{x_col} vs {y_col}")

plt.suptitle("Targeted Scatterplots for Strongest Relationships", y=1.03)
plt.tight_layout()
plt.show()
```



Correlation conclusions for modeling:

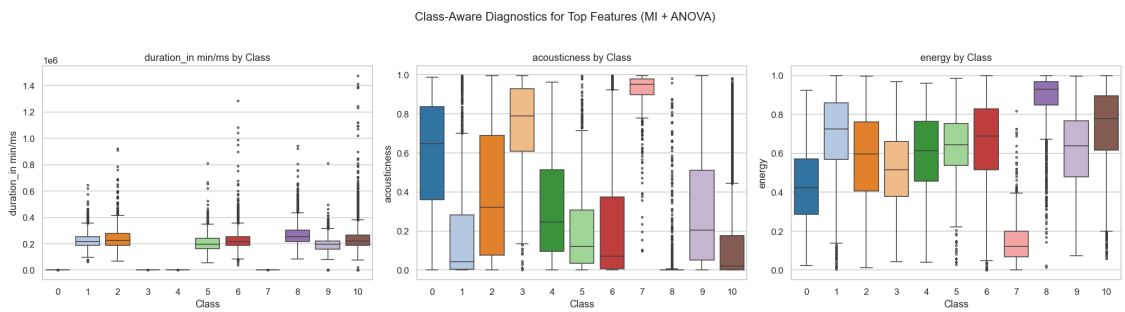
- `energy`, `loudness`, and `acousticness` form a correlated cluster (strong pairwise relationships), so they are partially redundant for linear models and can inflate coefficient variance without regularization.
- Redundancy does **not** mean immediate removal. We keep the cluster initially because each feature can still add complementary nonlinear signal, especially for tree-based models.
- Practical strategy: keep all three for baseline models, then use regularization and ablation tests to decide whether dropping one improves generalization and interpretability.

1.2.7 2.7. Class Separation in Top Signal Features (MI + ANOVA)

[46]:
```
# Question: Which top signal features differ most across classes?
# `Class` is categorical, so Pearson correlation to class codes is not␣
↪appropriate.
from sklearn.feature_selection import f_classif, mutual_info_classif

X = eda_df[FOCUSED_FEATURES].copy()
y = eda_df[TARGET_COL].copy()

# Handle missingness for score computation.
X_imputed = X.fillna(X.median(numeric_only=True))

f_vals, p_vals = f_classif(X_imputed, y)
mi_vals = mutual_info_classif(X_imputed, y, random_state=RANDOM_STATE)

signal_rank = pd.DataFrame({
    "anova_f": f_vals,
    "anova_pvalue": p_vals,
    "mutual_info": mi_vals,
}, index=FOCUSED_FEATURES).sort_values(["mutual_info", "anova_f"],␣
↪ascending=False)

display(signal_rank.round(4))

top3_by_class_signal = signal_rank.head(3).index.tolist()
print("Top 3 class-signal features (categorical-safe ranking):",␣
↪top3_by_class_signal)

fig, axes = plt.subplots(1, 3, figsize=FIGSIZE_THREE_PANEL, sharex=False)
for ax, feature in zip(axes, top3_by_class_signal):
    sns.boxplot(
        data=eda_df,
        x=TARGET_COL,
        y=feature,
        ax=ax,
        hue=TARGET_COL,
        order=class_order,
        palette=class_palette_map,
        fliersize=2,
        legend=False,
    )
    ax.set_title(f"{feature} by Class")
    ax.set_xlabel(TARGET_COL)

plt.suptitle("Class-Aware Diagnostics for Top Features (MI + ANOVA)", y=1.03)
plt.tight_layout()
```

```
plt.show()
```

```
                       anova_f  anova_pvalue  mutual_info
duration_in min/ms  1484.9658           0.0       0.3509
acousticness         947.4509           0.0       0.2774
energy               833.6812           0.0       0.2139
instrumentalness     551.6458           0.0       0.1718
speechiness          807.7890           0.0       0.1414
loudness             785.5009           0.0       0.1303
Popularity           203.9523           0.0       0.0978
tempo                 42.5693           0.0       0.0706

Top 3 class-signal features (categorical-safe ranking): ['duration_in min/ms',
'acousticness', 'energy']
```



- The class-conditional boxplots confirm that top-ranked features from MI + ANOVA (`duration_in min/ms`, `acousticness`, `energy`) show meaningful distribution shifts across classes rather than uniform overlap.
- Separation is present but not cleanly linear: several classes still overlap in central ranges, indicating that interaction effects and nonlinear boundaries are likely important for final performance.
- `duration_in min/ms` contributes strong class signal, but its broad spread/outlier behavior suggests this feature should be paired with robust preprocessing checks (scaling/transform sensitivity in linear-distance models).
- Because rank metrics and boxplots agree directionally, these features are justified for baseline inclusion; final keep/drop decisions should still be made through cross-validated ablation by model family.

[47]:
```
# Question: Is class imbalance large enough to affect model training and␣
↪evaluation?
class_counts = df[TARGET_COL].value_counts().sort_index()
class_pct = (class_counts / class_counts.sum() * 100).round(2)
imbalance_ratio = class_counts.max() / class_counts.min()

class_balance = pd.DataFrame({
    "count": class_counts,
```

```
    "pct": class_pct,
})

print(f"Imbalance ratio (largest/smallest class): {imbalance_ratio:.2f}x")
class_balance
```

```
Imbalance ratio (largest/smallest class): 12.79x
```

[47]:

	count	pct
Class		
0	625	3.47
1	1373	7.63
2	1272	7.07
3	402	2.23
4	387	2.15
5	1447	8.04
6	2587	14.38
7	576	3.20
8	1854	10.30
9	2524	14.03
10	4949	27.50

1.2.8 2.8. Modeling implications

1. **Imputation:** `instrumentalness` has meaningful missingness, so keep median imputation and test a missing-indicator feature to capture any missingness signal.
2. **Encoding choices:** treat `key` as categorical (not continuous) if included; drop it if cross-validation shows negligible incremental gain.
3. **Scale/transform policy:** use robust scaling for skewed heavy-tail variables (`speechiness`, `acousticness`, `duration_in min/ms`, and potentially `instrumentalness`), then test `log1p` as an ablation for linear/distance-based models.
4. **Correlation management:** keep `energy`/`loudness`/`acousticness` in baseline runs; manage redundancy with regularization and feature-drop ablations, especially for linear models.
5. **Class-aware training/evaluation:** class imbalance is substantial enough to affect learning, so prioritize macro-F1 and balanced accuracy, use stratified folds, and try class weights before oversampling.
6. **Data-quality constraints are currently stable:** no exact duplicates and no bounded-range/invalid-zero violations were found, so avoid unnecessary row/value deletion and preserve signal.
7. **Validation discipline:** compare audio-only versus metadata-augmented variants, and keep all preprocessing inside a single cross-validated pipeline so EDA insights are validated, not assumed.

1.3 3. Data Quality and Cleaning

This section documents the data-quality checks and cleaning actions used before modeling.

Run order: Execute cells from top to bottom. Run the setup cell first, then run sections 3.1 to 3.8 in sequence so each step uses the updated `data` object from prior steps.

- **Missing values**
  - **Issue:** Missingness is concentrated in a few columns, especially `instrumentalness`, with smaller gaps in `Popularity` and `key`.
  - **Change:** Impute missing values in `instrumentalness`, `Popularity`, and `key` using each column median.
  - **Why reasonable:** Median imputation is robust to skew and extreme values and preserves sample size.
  - **Model impact:** Keeps training rows that would otherwise be dropped; however, imputation can compress variance near the center, so this should be validated in cross-validation.
- **Duplicate records**
  - **Issue checked:** Exact duplicate rows were audited.
  - **Finding:** No exact duplicates were found in this dataset snapshot.
  - **Change:** Keep `drop_duplicates(keep="first")` as a defensive, reproducible safeguard, while noting that row count remains unchanged.
  - **Why reasonable:** Duplicate audits still protect pipeline reliability when data sources are refreshed.
  - **Model impact:** No effect in the current run; prevents accidental reweighting if duplicates appear later.
- **Invalid ranges and invalid zeros**
  - **Issue checked:** Bounded audio features (for example `energy`, `danceability`, `acousticness`, `instrumentalness`, `liveness`, `valence`, `speechiness`) should lie in `[0,1]`; variables like `tempo` and `duration_in min/ms` should be positive.
  - **Finding:** No bounded-range violations and no invalid zeros (`<= 0`) were detected in the audited columns.
  - **Change:** Preserve values as-is and retain these checks as automated guards.
  - **Why reasonable:** Constraint checks catch clearly invalid entries without arbitrary edits.
  - **Model impact:** Confirms current features are numerically plausible, reducing risk of scale distortion from corrupted inputs.
- **Outliers / extreme values**
  - **Issue:** Features such as `duration_in min/ms` and `tempo` can have heavy tails.
  - **Change:** Quantify outliers with an IQR-based diagnostic report; keep values unchanged in this notebook stage.
  - **Why reasonable:** For this dataset size, immediate trimming risks deleting valid rare songs; diagnostics-first is safer before model-based validation.
  - **Model impact:** Tree models are typically robust to extremes, while linear/distance models may need robust scaling or transforms in the modeling pipeline.
- **Inconsistent labels and clearly invalid entries**
  - **Issue checked:** Target label consistency and obvious invalid entries are reviewed via audits.
  - **Finding:** No missing target labels were found, and label cardinality is stable (`11` classes).
  - **Change:** Report findings explicitly; do not alter values unless a concrete rule is justified.
  - **Why reasonable:** Ensures transparency and reproducibility.
  - **Model impact:** Reduces hidden leakage/noise risks in downstream evaluation.
- **Feature inclusion decisions (preprocessing, not pure cleaning)**
  - `Artist Name` and `Track Name` are removed from this baseline because high-cardinality identifiers can cause leakage-like memorization and unstable generalization.

- `danceability` and `mode` are also removed in this baseline to limit early feature-space complexity; this is treated as a modeling choice to validate by ablation, not as a claim that these variables are inherently invalid.

Overall, the policy is conservative: correct clear quality issues, preserve plausible observations, and defer aggressive transformations/removals until cross-validated modeling confirms a benefit.

1.3.1 Setup

[48]:
```
data = pd.read_csv(TRAIN_PATH)
print(f"Rows before cleaning: {len(data):,}")
```

```
Rows before cleaning: 17,996
```

1.3.2 3.1 Missing Values Audit (Before Cleaning)

Check missingness in key columns before any transformations.

[49]:
```
missing_before = data[["instrumentalness", "Popularity", "key"]].isna().sum()
print("Missing values before imputation:")
print(missing_before)
```

```
Missing values before imputation:
instrumentalness    4377
Popularity           428
key                 2014
dtype: int64
```

1.3.3 3.2 Duplicate Audit and Removal

Count exact duplicates, remove them, and verify the post-cleaning row count.

[50]:
```
dup_before = data.duplicated().sum()
print(f"Exact duplicates before removal: {dup_before:,}")

data = data.drop_duplicates(keep="first").copy()

dup_after = data.duplicated().sum()
print(f"Exact duplicates after removal: {dup_after:,}")
print(f"Rows after duplicate removal: {len(data):,}")
```

```
Exact duplicates before removal: 0
Exact duplicates after removal: 0
Rows after duplicate removal: 17,996
```

1.3.4 3.3 Median Imputation for Selected Columns

Impute missing values in `instrumentalness`, `key`, and `Popularity` using the column median.

[51]:
```
impute_cols = ["instrumentalness", "key", "Popularity"]
for col in impute_cols:
    data[col] = data[col].fillna(data[col].median())

missing_after = data[impute_cols].isna().sum()
print("Missing values after imputation:")
print(missing_after)
```

```
Missing values after imputation:
instrumentalness    0
key                 0
Popularity          0
dtype: int64
```

1.3.5 3.4 Range Validity Checks for Bounded Features

Verify that bounded audio features are within `[0,1]`.

[52]:
```
range_violations = {
    col: int(((data[col] < 0) | (data[col] > 1)).sum())
    for col in BOUNDED_FEATURES_01 if col in data.columns
}

print("Range violations for [0,1] features:")
print(pd.Series(range_violations).sort_values(ascending=False))
```

```
Range violations for [0,1] features:
danceability        0
energy              0
speechiness         0
acousticness        0
instrumentalness    0
liveness            0
valence             0
dtype: int64
```

1.3.6 3.5 Invalid-Zero Checks

Check columns where values `<= 0` are implausible for observed measurements.

[53]:
```
invalid_zero_fields = ["tempo", "duration_in min/ms"]
invalid_zero_counts = {
    col: int((data[col] <= 0).sum())
    for col in invalid_zero_fields if col in data.columns
}

print("Invalid-zero checks (<= 0):")
print(pd.Series(invalid_zero_counts).sort_values(ascending=False))
```

```
Invalid-zero checks (<= 0):
tempo                 0
duration_in min/ms    0
dtype: int64
```

1.3.7 3.6 IQR Outlier Diagnostics

Compute IQR-based outlier counts for selected features (diagnostic only).

[54]:
```
outlier_features = ["tempo", "duration_in min/ms", "loudness", "Popularity",␣
 ↪"instrumentalness"]
iqr_report = []
for col in outlier_features:
    if col not in data.columns:
        continue
    q1 = data[col].quantile(0.25)
    q3 = data[col].quantile(0.75)
    iqr = q3 - q1
    lower = q1 - 1.5 * iqr
    upper = q3 + 1.5 * iqr
    outlier_count = int(((data[col] < lower) | (data[col] > upper)).sum())
    iqr_report.append({
        "feature": col,
        "q1": q1,
        "q3": q3,
        "iqr": iqr,
        "lower_bound": lower,
        "upper_bound": upper,
        "outlier_count": outlier_count,
        "outlier_pct": round(outlier_count / len(data) * 100, 3),
    })

print("IQR outlier diagnostic report (no values removed):")
print(pd.DataFrame(iqr_report).sort_values("outlier_pct", ascending=False))
```

```
IQR outlier diagnostic report (no values removed):
              feature             q1            q3           iqr  \
4    instrumentalness       0.000294       0.05660      0.056306
1  duration_in min/ms  166337.000000  252490.00000  86153.000000
2            loudness      -9.538000      -5.18900      4.349000
3          Popularity      33.000000      56.00000     23.000000
0               tempo      99.620750     141.96925     42.348500

    lower_bound    upper_bound  outlier_count  outlier_pct
4     -0.084165       0.141059           3714       20.638
1  37107.500000  381719.500000           3245       18.032
2    -16.061500       1.334500            818        4.545
3     -1.500000      90.500000             41        0.228
```

```
0      36.098000      205.492000              36         0.200
```

1.3.8 3.7 Target Label Sanity Checks

Check missing target labels and label cardinality.

[55]:
```
print("Target label checks:")
print(f"Missing Class labels: {int(data['Class'].isna().sum())}")
print(f"Unique Class labels: {int(data['Class'].nunique())}")
```

```
Target label checks:
Missing Class labels: 0
Unique Class labels: 11
```

1.3.9 3.8 Baseline Feature-Removal Snapshot (Modeling Choice)

Create the baseline modeling dataframe after conservative cleaning and selected column drops.

[56]:
```
baseline_drop_cols = ["Artist Name", "Track Name", "mode", "danceability"]
data_model = data.drop(columns=[c for c in baseline_drop_cols if c in data.
 ↪columns])

print("Columns dropped for baseline modeling:", baseline_drop_cols)
print(f"Rows retained for modeling: {len(data_model):,}")
print(f"Columns retained for modeling: {data_model.shape[1]}")
```

```
Columns dropped for baseline modeling: ['Artist Name', 'Track Name', 'mode',
'danceability']
Rows retained for modeling: 17,996
Columns retained for modeling: 13
```

1.4 4. Model Readiness Reflection

1.4.1 4.1. Most important patterns

- Genre prediction is feasible but non-trivial: classes overlap in several feature ranges, so purely linear boundaries are likely insufficient.
- `energy`, `loudness`, and `acousticness` form a correlated cluster; this indicates partial redundancy risk in linear models, but not automatic removal.
- Heavy-tail behavior is visible in features such as `duration_in min/ms`, `speechiness`, and `acousticness`, so preprocessing robustness matters.

1.4.2 4.2. Most important data quality issues

- Missingness is concentrated in a few columns (especially `instrumentalness`, with smaller gaps in `Popularity` and `key`), so imputation is preferred over dropping rows.
- Exact duplicate records should be removed to avoid overweighting repeated examples.
- Range/validity checks are necessary (bounded features in `[0,1]`, positive `tempo`/`duration_in min/ms`) before modeling.

1.4.3 4.3. Useful features (evidence-based)

- The strongest class-signal features from MI + ANOVA are `duration_in min/ms`, `acousticness`, and `energy`.
- `duration_in min/ms` appears especially important and should be kept in baseline models unless cross-validated ablation proves otherwise.
- `speechiness` is also meaningful and should be treated as a useful supporting feature.
- Feature usefulness must be confirmed by leakage-safe cross-validation and ablation, not EDA alone.

1.4.4 4.4. Preprocessing and modeling challenges

- Balancing signal retention vs redundancy: keep correlated features in baseline runs, then evaluate drop/keep via regularized models and ablation.
- Handling skew/outliers without deleting valid rare tracks: use robust scaling and test `log1p` sensitivity for linear/distance-based models.
- Managing class imbalance: prioritize macro-F1 and balanced accuracy, and use class-aware training strategies.
- Controlling leakage risk from metadata/high-cardinality identifiers: compare audio-only vs metadata-augmented models under the same CV protocol.

1.4.5 4.5. Final readiness statement

The dataset is ready for baseline modeling with a conservative pipeline: deduplicate, validate ranges, impute missing values, keep high-signal audio features (especially `duration_in min/ms`, `acousticness`, `energy`), and finalize feature decisions through cross-validated model comparison and ablation.